

Secure Network Design and Implementation for Tech Zolutions Inc.

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1 Introduction

A secure and well-structured network is essential for Tech Zolutions Inc., a growing software development company with 100 employees across three departments: Sales & Marketing (30 hosts), Development (50 hosts), and IT (20 hosts). Each department has specific network requirements, and the company also maintains a server room with a Web Server, DHCP Server, and Email Server, along with a conference room equipped with a video conferencing system.

This project focuses on designing and implementing a secure network that ensures efficient communication, resource allocation, and security. The implementation is divided into three tasks:

- **Network Design and Configuration** – Developing a network topology, performing subnetting, and configuring routers, switches, and DHCP.
- **Traffic Analysis** – Capturing and analysing network traffic to understand internal and inter-department communication.
- **Security Implementation** – Implementing access control measures, configuring a Zone-Policy Firewall (ZPF), Access Control List (ACL), and segmenting the network into zones of trust.

The final design will ensure secure, scalable, and reliable connectivity, meeting the company's operational needs while protecting critical resources.

2 Network Design and Configuration

2.1 Network Topology

The network is structured with a security router at the top, configured with a Zone-Policy Firewall (ZPF) using **INSIDE**, **OUTSIDE**, and **DMZ** zones to enforce secure traffic flow between internal departments and external access (Internet). This security router connects to a core router, which is linked via **dot1q** encapsulation to a Layer 2 core switch. The core switch distributes traffic to six access switches — serving the IT department, two switches for Sales & Marketing, one for the Conference Room, and three for Development.

Each department is segmented into its own VLAN and subnet as follows:

- VLAN 10 – Development
- VLAN 20 – Sales & Marketing
- VLAN 30 – IT
- VLAN 40 – Conference Room
- VLAN 50 – Server Room

The Network is divided into three zones:

- **DMZ**: Hosting the Web server (HTTP) accessible externally.

- **INSIDE:** Hosting Email and DHCP servers for internal communication and IP management.
- **OUTSIDE:** The Internet, which can only access resources within the DMZ.

The core router manages internal routing using the OSPF (Open Shortest Path First) protocol.

This design is optimised to be as cost-effective as possible, employing minimal components. Due to limitations in Cisco Packet Tracer, specifically the inability to apply ZPF to sub-interfaces, an additional router was used to set up ACL.

The overall architecture was selected to balance performance, security, scalability, and manageability. The hierarchical model, utilising a core router, a core switch, and multiple access switches, enhances network performance by reducing broadcast domains and isolating departmental traffic. VLAN segmentation improves security and fault isolation, while centralised routing simplifies network administration.

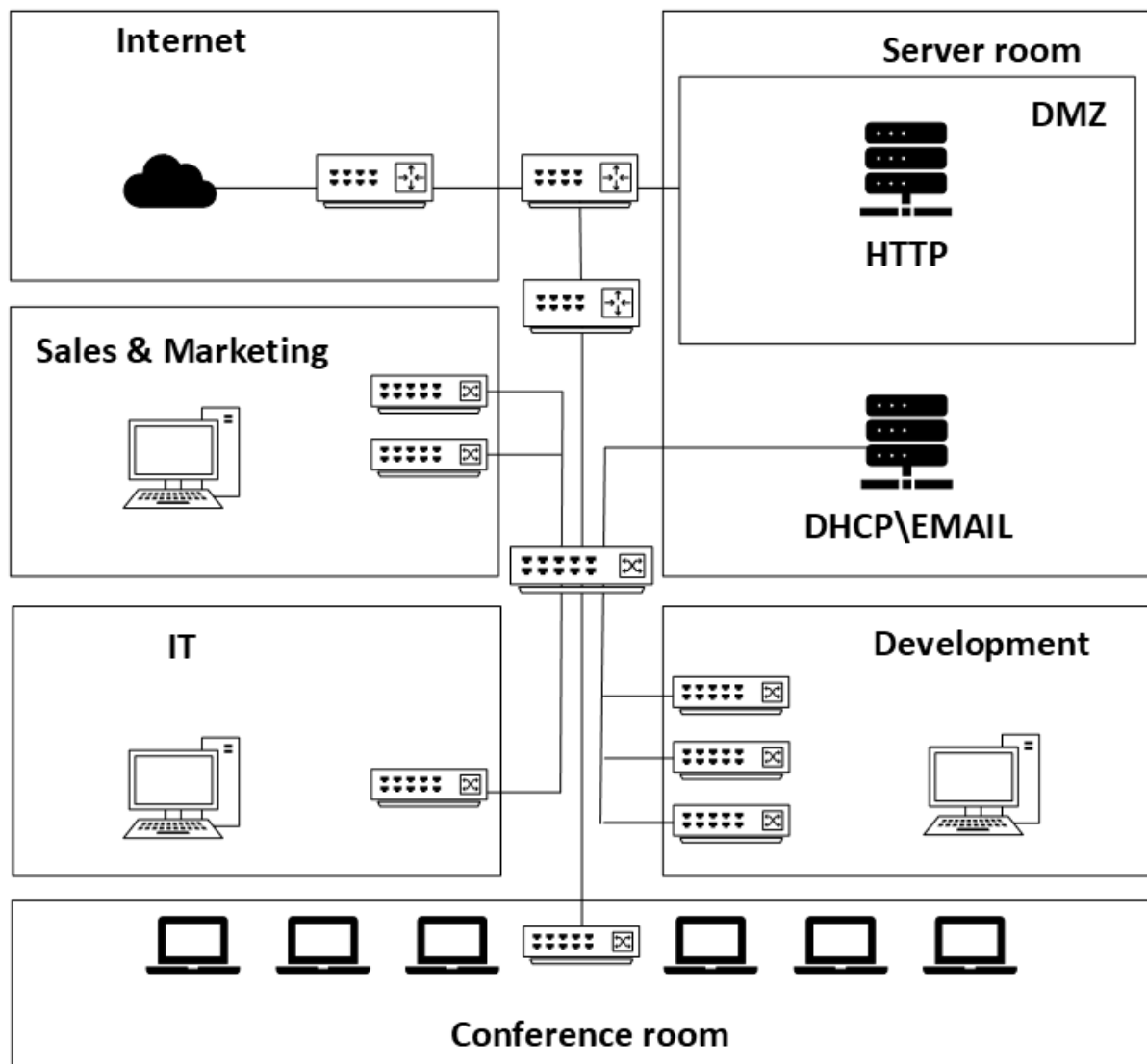


Figure 1: Network Design Overview

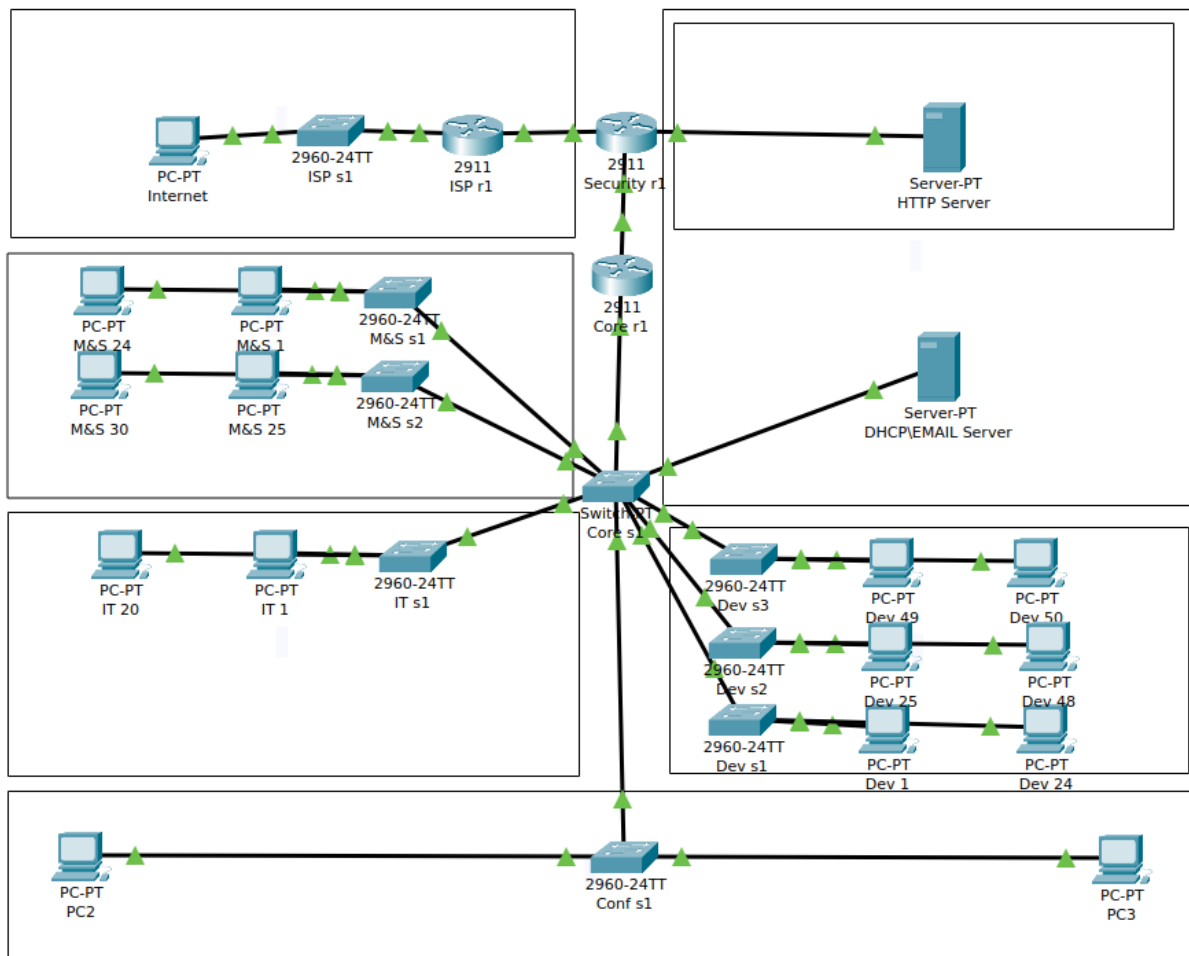


Figure 2: Final result in Cisco Packet Tracer

2.2 IP Addressing and Subnetting

First 3 bytes for every subnet are always the same and are 192.168.30

Department	IP Range	Subnet Mask	Devices	VLAN	Gateway
Development	.0 - .63	255.255.255.192 (/26)	50	10	.1
Sales & Marketing	.64 - .127	255.255.255.192 (/26)	30	20	.65
IT	.128 - .159	255.255.255.224 (/27)	20	30	.129
Conference Room	.160 - .175	255.255.255.240 (/28)	10	40	.161
Server Room	.176 - .179	255.255.255.252 (/30)	1	50	.177
Server Room DMZ	.180 - .183	255.255.255.252 (/30)	1	-	.181
Core to Security	.184 - .187	255.255.255.252 (/30)	1	-	-
ISP to Security	.188 - .191	255.255.255.252 (/30)	1	-	-
Internet	.192 - .195	255.255.255.252 (/30)	1	-	.193

Figure 3: Subnets of the network

2.3 Device Configuration

The first step at setting up our network is going to be to configure VLANs for each department.

First we are going to begin with secondary switches setup for each of them is pretty similar and can be done using the next template :

```
enable
conf t
hostname [Switch Name]
vlan [VLAN ID]
name [VLAN NAME]
int g0/1
switchport mode access
switchport access vlan [VLAN ID]
int range f0/1-24
switchport mode access
switchport access vlan [VLAN ID]
exit
exit
write memory
reload
```

Utilising this template to set up the switch in the Development department:

```
Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Dev_s1
Dev_s1(config)#vlan 10
Dev_s1(config-vlan)#name VLAN_10_DEV
Dev_s1(config-vlan)#int g0/1
Dev_s1(config-if)#switchport mode access
Dev_s1(config-if)#switchport access vlan 10
Dev_s1(config-if)#int range f0/1-24
Dev_s1(config-if-range)#switchport mode access
Dev_s1(config-if-range)#switchport access vlan 10
Dev_s1(config-if-range)#exit
Dev_s1(config)#exit
Dev_s1#write memory
Building configuration...
[OK]
Dev_s1#reload|
```

Figure 4: Setting up VLANs

This network uses the following naming conventions

VLAN Names:

VLAN names are written in uppercase and follow the format:

VLAN_[VLAN_ID]_[SHORT_DEPARTMENT_NAME]

Example: For VLAN 30 assigned to the IT department, the name is:

VLAN_30_IT

Device Names (Switches and Routers):

Device names follow the format:

[SHORT_DEPARTMENT_NAME]_[s/r] [DEVICE_ID]

Where:

- **s** stands for switch
- **r** stands for router

Examples:

- Core router: **Core_r1**
- Second switch in the Development department: **Dev_s2**

Next step is to set up a trunk port between the core switch and core router and connections between the core switch and secondary switches.

Before setting up ports, we need to create and name VLANs on the core switch.

```
Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name VLAN_10_DEV
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name VLAN_20_M&S
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name VLAN_30_IT
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name VLAN_40_CONF
Switch(config-vlan)#vlan 50
Switch(config-vlan)#name VLAN_50_SERVER
```

Figure 5: Generating VLANs on the core switch

Once all of the VLANs are created, we can set up the trunk port and VLANs on ports that are connected to secondary switches.

```
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allow vlan 10,20,30,40,50
```

Figure 6: Setting up trunk port

To speed up the configuration of the core switch, we can use the following template that can be copy and pasted:

```
int g[ID]/1
switchport mode access
switchport access vlan [VLAN ID]
```

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int g5/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
```

Figure 7: Setting up core switch access ports

Before configuring the router, we need to allocate a static IP address to the DHCP server.

☐ DHCP	☒ Static
IPv4 Address	192.168.30.178
Subnet Mask	255.255.255.252
Default Gateway	192.168.30.177

Figure 8: Allocating IP address to the DHCP server

Now it is time to set up the core router. Since multiple VLANs are connected to a single physical port on the router, we need to configure **Dot1Q** encapsulation to manage VLAN traffic. Additionally, we must set up the IP helper address to enable communication with the DHCP server, allowing clients in different VLANs to obtain IP addresses.

```

Core_r1(config-subif)#int g0/1.50
Core_r1(config-subif)#encapsulation dot1Q 50
Core_r1(config-subif)#ip address 192.168.30.177 255.255.255.252
Core_r1(config-subif)#int g0/1.40
Core_r1(config-subif)#encapsulation dot1Q 40
Core_r1(config-subif)#ip address 192.168.30.161 255.255.255.240
Core_r1(config-subif)#int g0/1.30
Core_r1(config-subif)#encapsulation dot1Q 30
Core_r1(config-subif)#ip address 192.168.30.129 255.255.255.224
Core_r1(config-subif)#int g0/1.20
Core_r1(config-subif)#encapsulation dot1Q 20
Core_r1(config-subif)#ip address 192.168.30.65 255.255.255.192
Core_r1(config-subif)#int g0/1.10
Core_r1(config-subif)#encapsulation dot1Q 10
Core_r1(config-subif)#ip address 192.168.30.1 255.255.255.192
Core_r1(config-subif)#

```

Figure 9: Configuring encapsulation

```

Core_r1(config-subif)#int g0/1.10
Core_r1(config-subif)#ip helper-address 192.168.30.178
Core_r1(config-subif)#int g0/1.20
Core_r1(config-subif)#ip helper-address 192.168.30.178
Core_r1(config-subif)#int g0/1.30
Core_r1(config-subif)#ip helper-address 192.168.30.178
Core_r1(config-subif)#int g0/1.40
Core_r1(config-subif)#ip helper-address 192.168.30.178
Core_r1(config-subif)#int g0/1.50
Core_r1(config-subif)#ip helper-address 192.168.30.178

```

Figure 10: Configuring IP helper address for correct work of DHCP server

Similar IP setup is done on **Security r1** and **ISP r1**

For the routers to communicate with each other, we need to set up **OSPF** (Open Shortest Path First), which can be done using the following configuration template:

```

enable
configure terminal
router ospf 1
router-id x.x.x.x
network [IP] [WILDCARD] area 0
end
write memory
reload

```

```

Core_r1(config)#router ospf 1
Core_r1(config-router)#router-id 1.1.1.1
Core_r1(config-router)#network 192.168.30.0 0.0.0.63 area 0
Core_r1(config-router)#network 192.168.30.64 0.0.0.63 area 0
Core_r1(config-router)#network 192.168.30.128 0.0.0.31 area 0
Core_r1(config-router)#network 192.168.30.160 0.0.0.15 area 0
Core_r1(config-router)#network 192.168.30.184 0.0.0.3 area 0

```

Figure 11: OSPF setup

Similar IP setup is done on **Security r1** and **ISP r1**

To make our network fully functional, we need to add DHCP IP pools so that computers can obtain their IP addresses automatically.

DHCP

Interface: GigabitEthernet0

Service: ☒ On ☐ Off

Pool Name: Development 10
 Default Gateway: 192.168.30.1
 DNS Server: 0.0.0.0
 Start IP Address: 192 168 30 2
 Subnet Mask: 255 255 255 192
 Maximum Number of Users: 61
 TFTP Server: 0.0.0.0
 WLC Address: 0.0.0.0

Add
Save
Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Conference room 40	192.16...	0.0.0.0	192.16...	255.25...	13	0.0.0.0	0.0.0.0
IT 30	192.16...	0.0.0.0	192.16...	255.25...	29	0.0.0.0	0.0.0.0
Sales & Marketing 20	192.16...	0.0.0.0	192.16...	255.25...	29	0.0.0.0	0.0.0.0
Development 10	192.16...	0.0.0.0	192.16...	255.25...	61	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	0.0.0.0	0	0.0.0.0	0.0.0.0

Figure 12: Example of a DHCP pool

The setup of the **HTTP** and **Email** servers is fairly straightforward.

The HTTP server works out of the box and does not require any additional configuration.

The **SMTP** and **POP3** services are also enabled by default; the only task remaining is to add users.

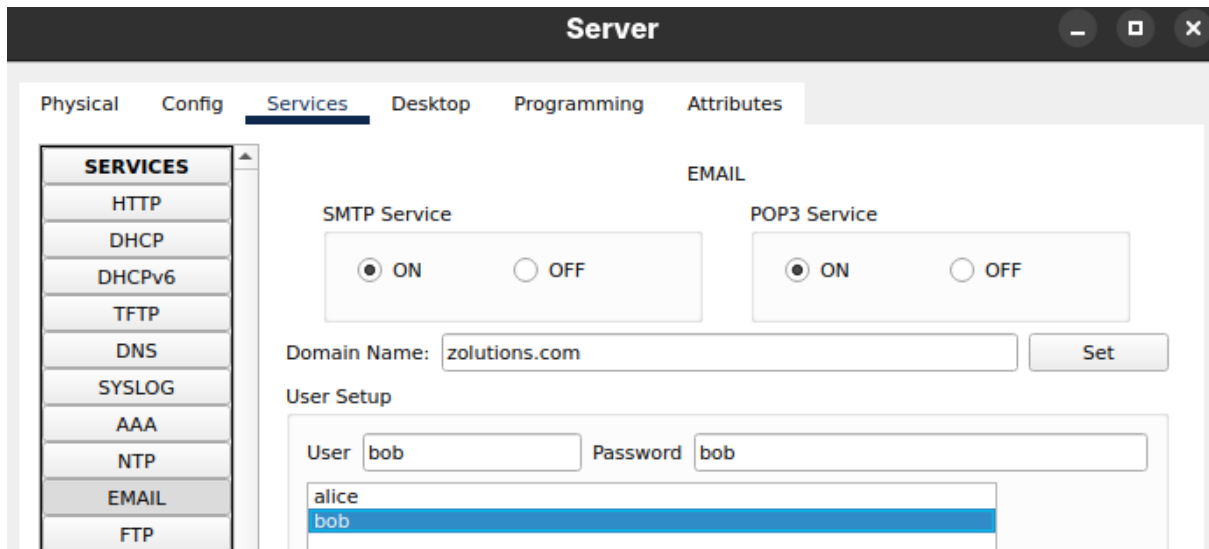


Figure 13: Setting up EMAIL server

2.4 Testing and Verification

3 Traffic Analysis

3.1 Network Traffic Capture & Protocol Analysis

To better understand different protocols and network behaviour, I am going to analyse ping (ICMP) and HTTP packets.

When testing connectivity, the `ping` utility is commonly used. Let us analyse the behaviour of the network when pinging a host with the IP address 192.168.30.5 from another host with the IP address 192.168.30.7, both within the same VLAN.

We can easily visualise network traffic using Cisco's Simulation Mode, which allows us to observe packet data and track how packets move through the network.

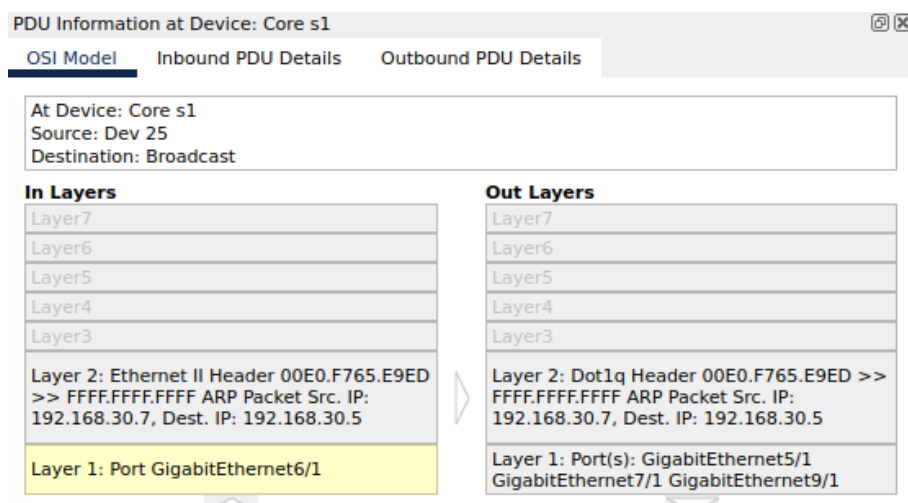


Figure 14: ARP request

The first packet sent into the network is not an ICMP packet, but an ARP (Address Resolution Protocol) packet. When devices communicate on a local network, they use MAC (Media Access Control) addresses to send data directly to a device's port.

As seen in the captured traffic, the computer with the IP address **192.168.30.7** sends an ARP request to the broadcast MAC address **FF:FF:FF:FF:FF:FF**, which essentially says:

"Hey, who has IP 192.168.30.5? Tell me your MAC address."

The sender will use the response from this packet to update its ARP table, which stores IP-to-MAC address mappings.

It is important to note that ARP has certain security weaknesses, as it lacks authentication mechanisms. This allows attackers to send fake ARP messages and poison a network's ARP cache. Such vulnerabilities enable man-in-the-middle (MITM) attacks, where an attacker can intercept and manipulate communications between devices by redirecting traffic through their own system.

After the ARP table is updated, the first device is ready to send ICMP packets to initiate the ping operation towards the second device.

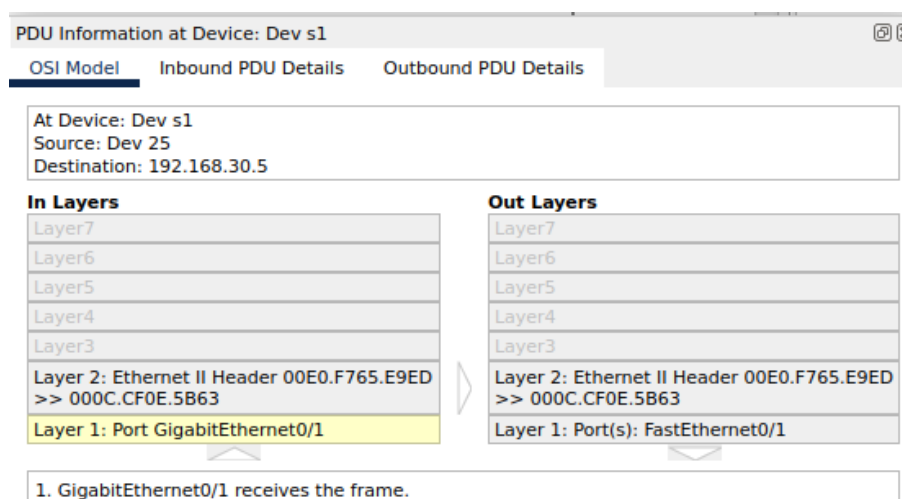


Figure 15: ICMP packet

As we can see all of the pings to the 192.168.30.5 were successful.

```
C:\>ping 192.168.30.5

Pinging 192.168.30.5 with 32 bytes of data:

Reply from 192.168.30.5: bytes=32 time=16ms TTL=128
Reply from 192.168.30.5: bytes=32 time=8ms TTL=128
Reply from 192.168.30.5: bytes=32 time=9ms TTL=128
Reply from 192.168.30.5: bytes=32 time=8ms TTL=128

Ping statistics for 192.168.30.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 16ms, Average = 10ms
```

Figure 16: ARP

Now let's see how the network behaves when it comes to pings across vlans

PDU Information at Device: Core r1

OSI Model Inbound PDU Details Outbound PDU Details

At Device: Core r1
Source: Dev 25
Destination: Broadcast

In Layers

Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Dot1q Header 00E0.F765.E9ED >> FFFF.FFFF.FFFF ARP Packet Src. IP: 192.168.30.7, Dest. IP: 192.168.30.1
Layer 1: Port GigabitEthernet0/1

Out Layers

Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Dot1q Header 0040.0B78.1902 >> 00E0.F765.E9ED ARP Packet Src. IP: 192.168.30.1, Dest. IP: 192.168.30.7
Layer 1: Port(s): GigabitEthernet0/1

1. GigabitEthernet0/1 receives the frame.

Figure 17: ICMP

When pinging 192.168.30.68 from 192.168.30.7, the destination IP address lies outside the local network's IP range. As a result, an ARP request is sent to the router to obtain its MAC address, allowing the ICMP packet to be forwarded to the default gateway.

PDU Information at Device: Core r1

OSI Model Inbound PDU Details Outbound PDU Details

At Device: Core r1
Source: Dev 25
Destination: 192.168.30.68

In Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.30.7, Dest. IP: 192.168.30.68 ICMP Message Type: 8
Layer 2: Dot1q Header 00E0.F765.E9ED >> 0040.0B78.1902
Layer 1: Port GigabitEthernet0/1

Out Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.30.7, Dest. IP: 192.168.30.68 ICMP Message Type: 8
Layer 2:
Layer1

1. GigabitEthernet0/1 receives the frame.

Figure 18: ICMP

After the ARP table is updated, the ping is sent to the router, which then sends an ARP request within its network to obtain the destination's MAC address.

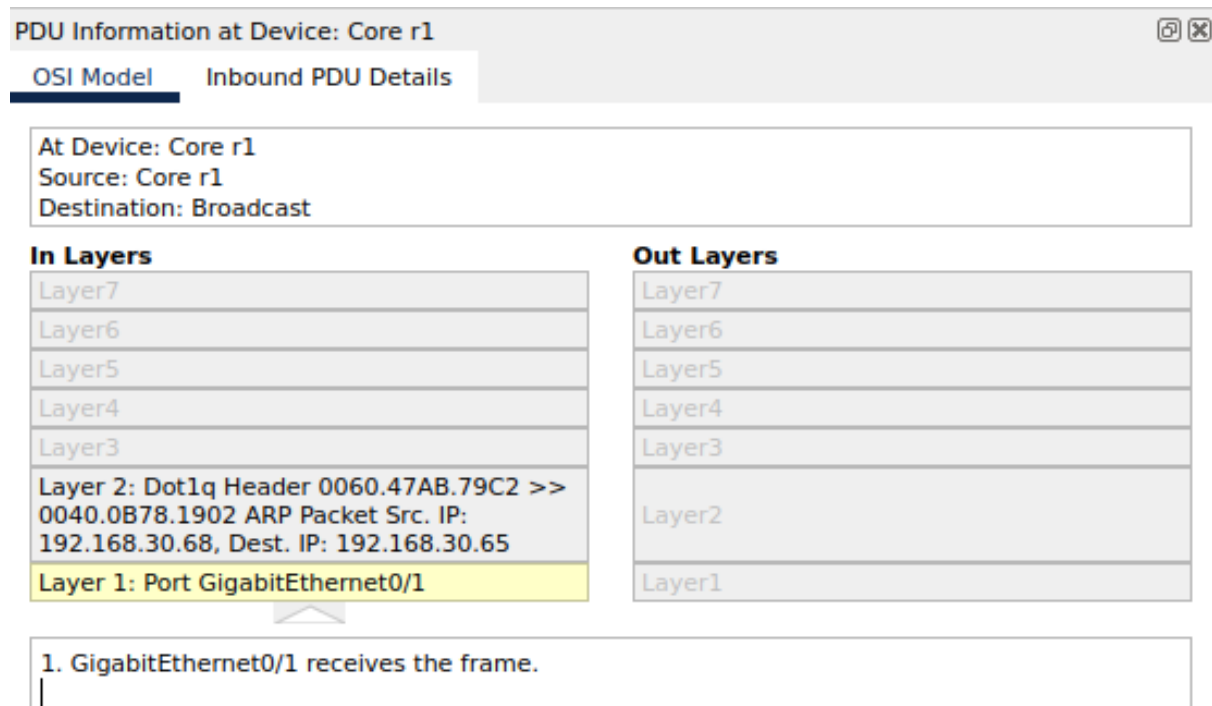


Figure 19: ARP

Since the router ignores the first packet because the destination device's MAC address is not in its ARP table, the initial ping times out. The second ping is then sent after the ARP table is updated.

```
C:\>ping 192.168.30.68

Pinging 192.168.30.68 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.68: bytes=32 time<1ms TTL=127
Reply from 192.168.30.68: bytes=32 time<1ms TTL=127
Reply from 192.168.30.68: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.68:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 20: Successful ping between departments

Computers from different departments access an HTTP server using the TCP protocol to establish and maintain a reliable connection through a three-way handshake. Once the connection is established, an HTTP request is sent to the server, which responds with an HTTP packet containing the website's data.

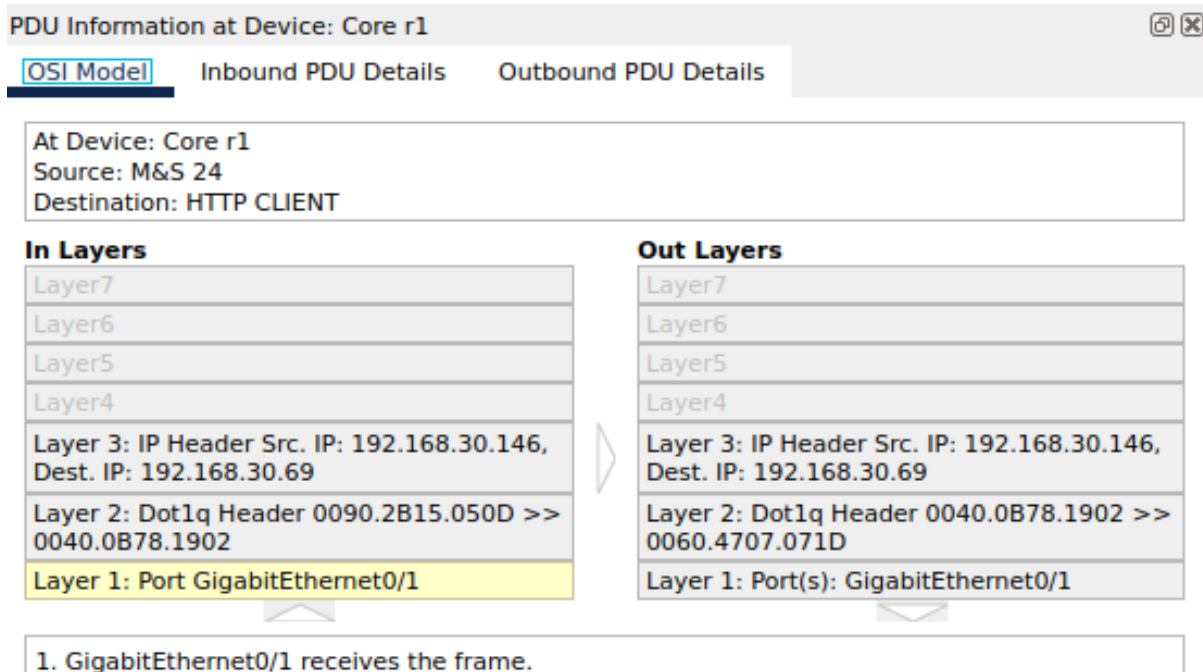


Figure 21: HTTP traffic

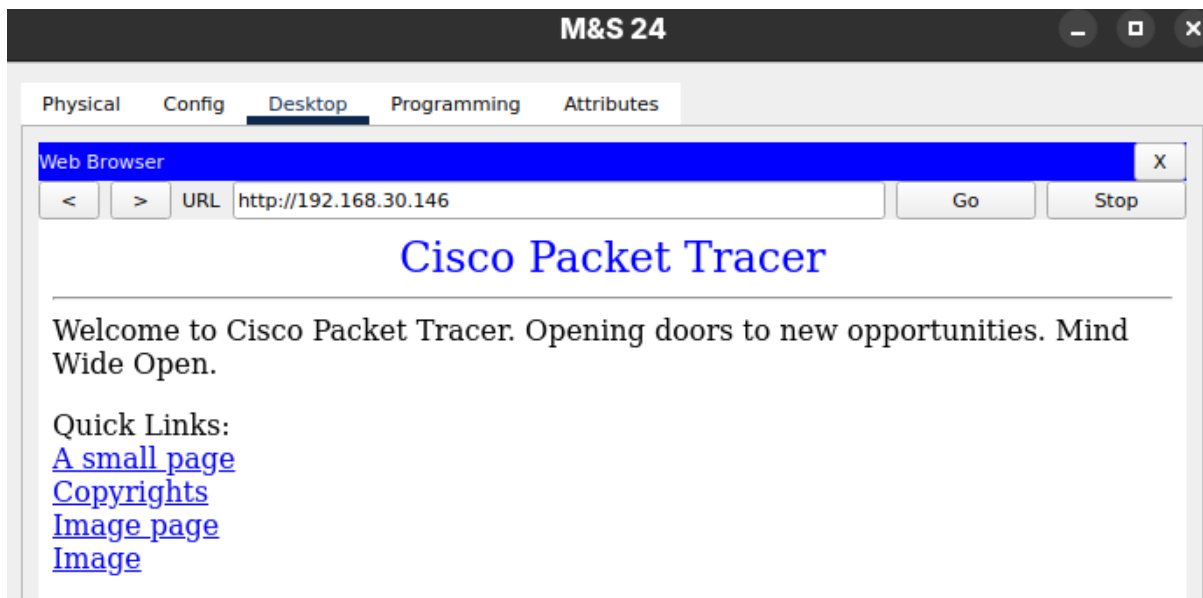


Figure 22: Website accessed successfully

4 Security and Zones of Trust

4.1 Security Configurations

No network can be considered truly secure if administrative access to networking devices is left unprotected. To ensure that only authorised personnel, typically network

administrators, can modify security configurations, it is essential to implement password protection at various levels. The following template is applied to every network device to set a baseline level of access security by setting console and enable mode passwords, and encrypting all plaintext passwords:

```
enable
configure terminal
no ip domain-lookup
banner motd #
+++ ACCESS RESTRICTED +++
Only the faithful may proceed.
Unauthorised access will be purged with righteous fire.
BOW BEFORE THE MACHINE GOD! The flesh is weak;
the Omnissiah's will is eternal.
+++ GLORY TO THE OMNISSIAH +++
#
line console 0
password Zero1234!
login
exit
enable secret ZeroOne234!
exit
exit
Zero1234!

enable
ZeroOne234!

conf t
service password-encryption

write
reload
```

- Implementation of strong passwords and encryption - Justification of security measures

4.2 Network Segmentation and Trust Zones

It is important to prevent any unauthorised traffic from accessing the internal network. However, we still need to provide customers with access to the company's website.

To achieve this, we will configure three security zones:

DMZ, **INSIDE**, and **OUTSIDE**.

The **DMZ** (Demilitarised Zone) will host publicly accessible services, such as the web server, while the **INSIDE** zone will contain the internal, secure resources. The **OUTSIDE** zone represents the Internet. Traffic between these zones will be tightly controlled using firewall rules to ensure both accessibility and security.

We are going to configure access control between zones with the following rules:

- Only **HTTP** and **HTTPS** traffic will be allowed from the **OUTSIDE**, and **INSIDE** zones to the **DMZ** zone.
(*Note: No direct access from OUTSIDE to INSIDE is permitted.*)
- From the **INSIDE** zone to the **OUTSIDE** zone, only the following protocols will be allowed:
 - **HTTP**
 - **HTTPS**
 - **DNS**
 - **ICMP**

These rules ensure that public access is restricted strictly to web services hosted in the DMZ, while internal users maintain controlled access to external resources. This setup enforces strong security boundaries while preserving essential functionality.

Within the internal zone, each department (Development, Sales & Marketing, and Conference Room) is granted access to the servers located in the Server Room, and the Internet, but inter-department communication is restricted. The IT department, however, is accessible to all other departments and is also permitted to access any department in return.

4.3 Firewall and Access Control Policies

Initially, the **Cisco 2911 router** does not support Zone-Policy Firewalls (ZPF) by default. To enable this functionality, we must activate the appropriate security feature using the following command:

```
license boot module c2900 technology-package securityk9
```

After enabling the module, a router **reboot** is required for the changes to take effect.

Once the router has restarted, we can proceed to create the necessary **security zones**, which are essential for defining and controlling traffic flows between different parts of the network.

```
Router(config)#zone security OUTSIDE
Router(config-sec-zone)#zone security INSIDE
Router(config-sec-zone)#zone security DMZ
```

Figure 23: Create Zones of trust

```

Router(config-sec-zone)#int g0/2
Router(config-if)#zone-member security DMZ
Router(config-if)#int g0/1
Router(config-if)#zone-member security INSIDE
Router(config-if)#int g0/0
Router(config-if)#zone-member security OUTSIDE

```

Figure 24: Assign zones to interfaces

When zones are assigned to the interfaces, we need to create access maps

```

Router(config)#class-map type inspect match-any DMZ-SERVICES
Router(config-cmap)# match protocol http
Router(config-cmap)# match protocol https
Router(config-cmap)#
Router(config-cmap)#class-map type inspect match-any INSIDE-OUT
Router(config-cmap)# match protocol http
Router(config-cmap)# match protocol https
Router(config-cmap)# match protocol dns
Router(config-cmap)# match protocol icmp
Router(config-cmap)#

```

Figure 25: Create Access Maps

After access maps are created we need to put them under policy maps.

```

Router(config-pmap-c)#policy-map type inspect INSIDE-TO-OUTSIDE
Router(config-pmap)# class type inspect INSIDE-OUT
Router(config-pmap-c)# inspect
Router(config-pmap-c)# class class-default
Router(config-pmap-c)# drop

```

Figure 26: Policy map

The final step in the Zone-Policy Firewall (ZPF) configuration is to create a **zone pair**. Zone pairs are used to inform the router which **policy map** should be applied to traffic flowing between two specific zones while defining both the **source** and **destination** zones.

This configuration step allows the router to apply the correct inspection or filtering rules to traffic, based on the direction and the type of service being accessed.

```

Core_r1(config)#ip access-list extended ACL-DEV
Core_r1(config-ext-nacl)# ! M&S
Core_r1(config-ext-nacl)# deny ip any 192.168.30.64 0.0.0.63
Core_r1(config-ext-nacl)# ! Conf
Core_r1(config-ext-nacl)# deny ip any 192.168.30.160 0.0.0.15
Core_r1(config-ext-nacl)#
Core_r1(config-ext-nacl)# ! Allow all other traffic
Core_r1(config-ext-nacl)# permit ip any any
Core_r1(config-ext-nacl)#
Core_r1(config-ext-nacl)# interface GigabitEthernet0/1.10
Core_r1(config-subif)# ip access-group ACL-DEV in

```

Figure 28: Creating then assigning ACL for the Development Department

```

Router(config-pmap-c)#zone-pair security OUTSIDE-DMZ source OUTSIDE destination DMZ
Router(config-sec-zone-pair)# service-policy type inspect OUTSIDE-TO-DMZ
Router(config-sec-zone-pair)#
Router(config-sec-zone-pair)#zone-pair security INSIDE-OUTSIDE source INSIDE destination
OUTSIDE
Router(config-sec-zone-pair)# service-policy type inspect INSIDE-TO-OUTSIDE
Router(config-sec-zone-pair)#
Router(config-sec-zone-pair)#zone-pair security INSIDE-DMZ source INSIDE destination DMZ
Router(config-sec-zone-pair)# service-policy type inspect INSIDE-TO-DMZ
Router(config-sec-zone-pair)#

```

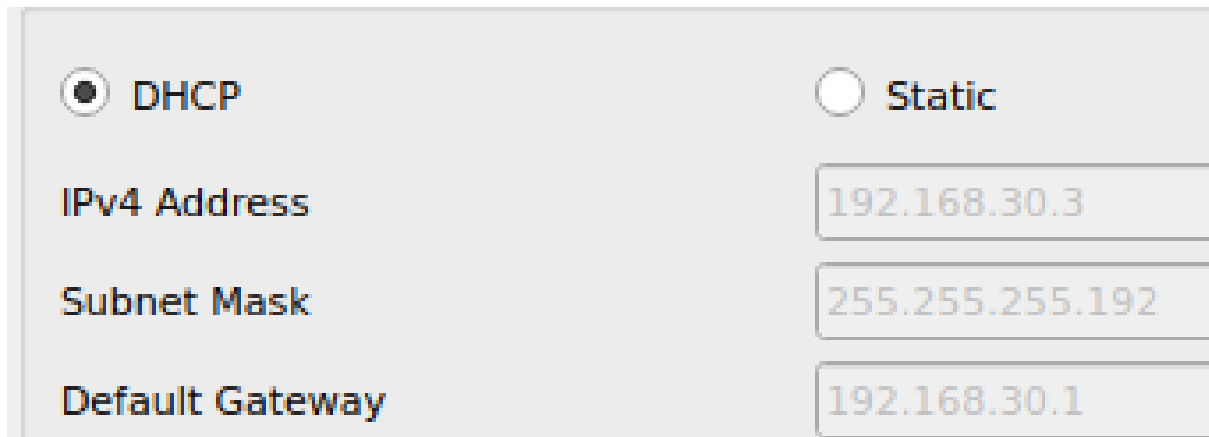
Figure 27: Zone pairs

If a ZPF is like airport security — managing how people move between different zones of the terminal (public, check-in, secure areas) — then an Access Control List (ACL) is more like a bouncer at every individual door, meticulously checking your pass before letting you through.

The setup process for an ACL is similar to that of a ZPF. You first define the access policy, and then assign it to the appropriate interface to enforce it.

The ACL explicitly denies traffic to each departmental subnet, followed by a general permit rule to allow all remaining traffic.

4.4 Testing and Verification



The image shows a configuration window for a network device. At the top, there are two radio buttons: 'DHCP' (which is selected) and 'Static'. Below these, there are three input fields. The first is labeled 'IPv4 Address' and contains the value '192.168.30.3'. The second is labeled 'Subnet Mask' and contains the value '255.255.255.192'. The third is labeled 'Default Gateway' and contains the value '192.168.30.1'.

Figure 29: Successful communication between devices on the INSIDE zone and the DHCP server

4.4.1 Testing OUTSIDE and INSIDE to DMZ connection

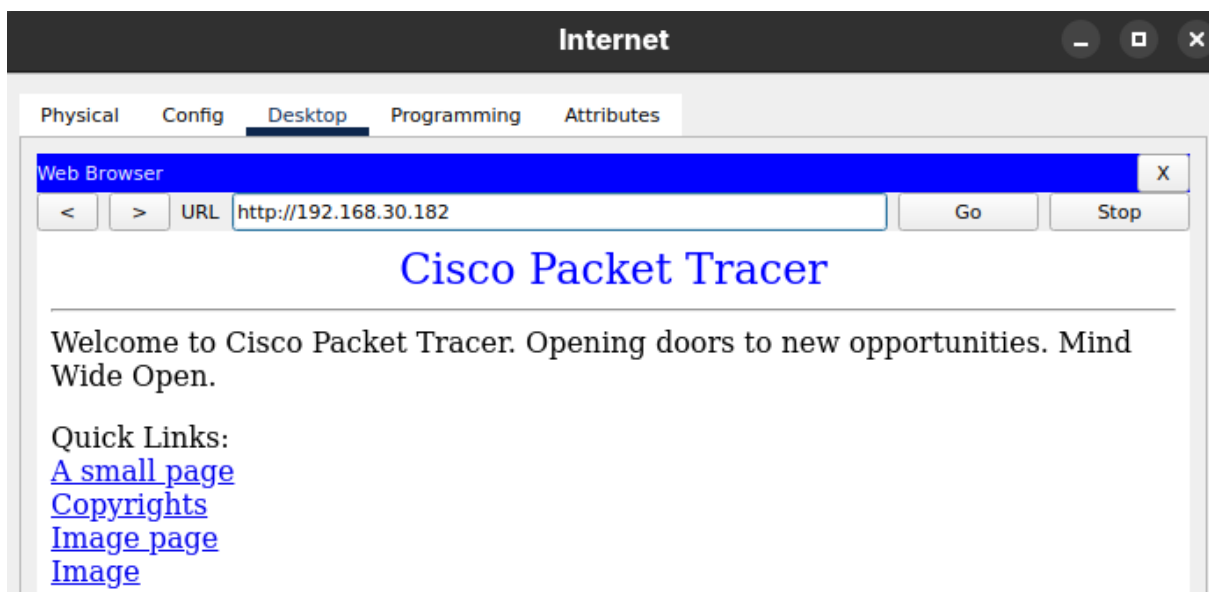


Figure 30: Successfully accessed HTTP from the internet



Figure 31: INSIDE zone can access HTTP server

```
C:\>ping 192.168.30.182

Pinging 192.168.30.182 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.182:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 32: ICMP packets from Internet to the DMZ were blocked

4.4.2 Testing OUTSIDE to INSIDE connection

```
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 33: Ping packets are successfully dropped from outside to inside zones

4.4.3 Testing INSIDE to OUTSIDE connection

```
C:\>ping 192.168.30.194

Pinging 192.168.30.194 with 32 bytes of data:

Reply from 192.168.30.194: bytes=32 time<1ms TTL=125
Reply from 192.168.30.194: bytes=32 time<1ms TTL=125
Reply from 192.168.30.194: bytes=32 time<1ms TTL=125
Reply from 192.168.30.194: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.30.194:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 34: Successful ping from from inside to outside zones

4.4.4 Testing connection between departments

```
C:\>ping 192.168.30.69

Pinging 192.168.30.69 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.30.69:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 35: Blocked pings from the development to the Sales & Marketing

```
C:\>ping 192.168.30.4

Pinging 192.168.30.4 with 32 bytes of data:

Reply from 192.168.30.4: bytes=32 time<1ms TTL=127
Reply from 192.168.30.4: bytes=32 time<1ms TTL=127
Reply from 192.168.30.4: bytes=32 time<1ms TTL=127
Reply from 192.168.30.4: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 36: Successful ping from IT to the Development

5 Conclusion

This project successfully designed and implemented a secure and efficient network for Tech Zolutions Inc. by utilising VLANs, Zone-Policy Firewalls (ZPF), and Access Control Lists (ACL) to ensure security. The network's scalability was achieved through OSPF routing, and performance was optimised with core and access switches. Security issues, like plaintext passwords, were addressed, ensuring a reliable and secure network. Overall, the design meets the company's current needs while providing a foundation for future growth and adaptability.